TASK: 1

Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Aim: To block all outgoing network traffic from the Spotify desktop application using Windows Defender Firewall.

Steps:

1. Open the **Start Menu** and search for **Windows Defender Firewall with Advanced Security**.
2. Click **Outbound Rules** in the left panel.
3. Select **New Rule** from the right panel.
4. Choose **Program** and click **Next**.
5. Select **This program path** and browse to the Spotify executable, usually `Spotify.exe`. Its location depends on how Spotify was installed.
6. Click **Next** and select **Block the connection**.
7. Select all three profiles: **Domain, Private, and Public**.
8. Click **Next** and enter the rule name as **Block Spotify Outbound Traffic**.
9. Click **Finish** to create the rule.

Result: The outbound rule blocks network connections initiated by the specified Spotify executable when the rule applies. Other Spotify components or executables may require separate rules.

TASK: 2

Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Aim: To create an inbound firewall rule for TCP port 443 associated with Google Chrome.

Steps:

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules** in the left panel.
3. Select **New Rule**.
4. Choose **Port** and click **Next**.
5. Select **TCP** and choose **Specific local ports**.
6. Enter `443` and click **Next**.
7. Select **Allow the connection** and click **Next**.
8. Select the required network profiles: Domain, Private, and Public, as appropriate.
9. Name the rule **Allow Chrome HTTPS Inbound TCP 443**.
10. Click **Finish**.

Important clarification: This rule allows matching inbound TCP connections on local port 443, but it does not restrict them to Google Chrome. To associate the rule with Chrome, choose **Program** instead of **Port**, specify Chrome's executable path, and configure the required port and protocol restrictions using the rule's properties. Also, Chrome normally initiates outbound HTTPS connections, so an inbound rule is not required for ordinary Chrome browsing.

Result: The inbound rule is created for TCP port 443. Its actual effect depends on the existing firewall rules and the network traffic being received.

TASK: 3

Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

Aim: To block incoming network traffic from the specified IP address using the Scope tab.

Steps:

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules** in the left panel.
3. Click **New Rule** in the right panel.
4. Select **Custom** and click **Next**.
5. Select **All programs** and click **Next**.
6. Leave the protocol setting as **Any**, unless a specific protocol is required.
7. On the **Scope** page, find **Remote IP address**.
8. Select **These IP addresses** and click **Add**.
9. Enter `123.45.67.89` and click **OK**.
10. Click **Next** and select **Block the connection**.
11. Select all applicable profiles: Domain, Private, and Public.
12. Name the rule **Block IP 123.45.67.89**.
13. Click **Finish** to save the rule.

Result: The rule blocks matching inbound traffic from `123.45.67.89` across the selected profiles. To block traffic in both directions, create a corresponding outbound rule and specify the address under the appropriate remote IP scope.

TASK: 4

Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Feature	Hardware Firewall	Software Firewall
1. Installation	A physical device installed between a network and its internet connection.	A program or operating-system feature installed on an individual computer or server.
2. Protection	Can protect multiple devices across an entire network.	Primarily controls network traffic to and from the device where it runs.
3. Cost and management	Often involves additional equipment, licensing, and network administration.	Often included with the operating system, such as Windows Defender Firewall, and managed on each device.
Use case	A large office uses a hardware firewall to protect its network and apply centralized security policies.	A student uses Windows Defender Firewall to block an application or restrict unwanted connections on a laptop.